Logarithms of Fibonacci Numbers

Mingkai Liu

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Abstract

The exponential growth of Fibonacci numbers becomes almost affine after taking logarithms. This paper investigates the consequences of that observation for ratios of consecutive logarithms. Starting from Binet's formula, we prove the uniform inequality

$$\frac{n-2}{n-1} < \frac{\log F_n}{\log F_{n+1}} < \frac{n-1}{n} \qquad (n \geq 3),$$

under the convention $F_0 = 0$, $F_1 = 1$. The proof separates a positive constant margin from an explicitly bounded, alternating correction. The inequality yields monotonicity of the logarithmic ratios, strict log-concavity of $(\log F_n)$, and two root sequences converging to the golden ratio from opposite sides. We introduce an effective shift, determine its global extrema, and obtain the optimal real constant shifts for a uniform rational enclosure. An exact remainder formula describes the parity of the approximation error and its exponential decay, while an asymptotic expansion identifies the first two algebraic orders of convergence. We extend the argument to nonadjacent indices and to positive sequences with a dominant exponential term. Finally, the same logarithmic decomposition yields a proof of Benford's law in every integer base and an explicit example of dependence between consecutive leading digits. Computations use exact Fibonacci integers and arbitrary-precision logarithms; code, data, and figures accompany the paper. The work combines a literature review with self-contained derivations and does not claim that the results presented here are previously unpublished.

Keywords: Fibonacci numbers; Binet's formula; logarithmic ratios; asymptotic expansion; monotonicity; Benford's law.
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1 Introduction and research context

The Fibonacci sequence is defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \geq 0). \tag{1.1}$$

Its first positive terms are $1, 1, 2, 3, 5, 8, 13, \ldots$. The familiar quotient F_{n+1}/F_n approaches the golden ratio, but the logarithmic quotient

$$R_n = \frac{\log F_n}{\log F_{n+1}}, \qquad n \geq 3, \tag{1.2}$$

approaches 1. The latter quotient measures the relative growth of the logarithmic scale of the sequence. It is also the logarithm of F_n to base F_{n+1}, and therefore expresses the exponent needed to obtain F_n from F_{n+1}.

The motivating question concerns a pair of rational bounds whose numerators and denominators differ by one. Under convention (1.1), we shall prove

$$\frac{n-2}{n-1} < R_n < \frac{n-1}{n} \qquad (n \geq 3). \tag{1.3}$$

The indexing convention matters. If instead $G_1 = 0$, $G_2 = 1$, then $G_n = F_{n-1}$ and (1.3) becomes

$$\frac{n-3}{n-2} < \frac{\log G_n}{\log G_{n+1}} < \frac{n-2}{n-1} \qquad (n \geq 4). \tag{1.4}$$

Thus the originally proposed form is valid after a one-place shift. With $F_1 = F_2 = 1$, its proposed upper bound fails already at $n = 3$, since $\log 2/\log 3 > 1/2$. An indexing correction is essential to the theorem, rather than a change to the underlying argument.

Our investigation has three main questions. First, why should the integers 1 and 2 occur as shifts in (1.3)? Second, how can an asymptotic explanation be converted into a statement valid at every admissible index? Third, what other information is contained in the same logarithmic representation? These questions lead naturally to explicit remainders, root monotonicity, sharp constant shifts, and the distribution of leading digits.

All unqualified logarithms in this paper are natural logarithms. The quotient (1.2) is unchanged by using any common logarithmic base $B > 0$, $B \neq 1$, because the factor $1/\log B$ cancels. For inequalities involving individual logarithms we use the natural base, and for leading digits we specify the numeral base explicitly. The exclusion of $n = 1, 2$ avoids a zero denominator at $n = 1$ and the failure of strictness at $n = 2$.

1.1 Connections with existing literature

There are two relevant strands of literature. The first studies monotonicity and concavity after normalizing combinatorial sequences by roots. Hou, Sun, and Wen [3] prove that the adjacent quotient of the root sequence $F_n^{1/n}$ decreases for $n \geq 4$, and consider more general Lucas sequences. Xia and Zhang [4] develop methods for establishing log-concavity of root sequences. These works concern a different normalization from R_n, but they share the principle that taking logarithms converts multiplicative assertions into manageable differences.

The second strand studies fractional parts of logarithms. Duncan [1] uses uniform distribution of Fibonacci logarithms in a problem related to digit counts and the Euclidean algorithm. Washington [2] proves Benford's law for Fibonacci and Lucas numbers in arbitrary integer bases. Weyl's criterion [5] supplies the equidistribution mechanism. Berger and Hill [6] organize the wider theory of significant-digit distributions. Cai, Hildebrand, and Li [7] distinguish global digit frequencies from the stronger requirement of local independence. These distinctions will be visible in a simple Fibonacci digit-pair computation.

A third, deeper application of logarithms appears in exponential Diophantine equations. Bugeaud, Mignotte, and Siksek [8] combine linear forms in logarithms with modular methods to classify perfect powers in the Fibonacci and Lucas sequences. Our arguments do not reproduce that machinery. We explain only the elementary logarithmic expression that connects the present estimates to such problems.

The contribution of this paper is the organization of these connections around one exact model and the provision of complete proofs and reproducible calculations for the claims we develop. A targeted literature search is not a certification of novelty. In particular, the bounds and sharp-shift formulation below are presented as results derived here, without a priority claim. The reader can assess them independently of any assumption that they are new.

2 Binet's formula and an exact logarithmic model

Let

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2} = -\phi^{-1}, \qquad q = \phi^{-2} = \frac{3-\sqrt{5}}{2}. \tag{2.1}$$

The numbers ϕ and ψ are the roots of $x^2 - x - 1 = 0$. Consequently the sequence $(\phi^n - \psi^n)/\sqrt{5}$ satisfies (1.1) and has its prescribed initial values. Uniqueness of a recurrence sequence with fixed initial values proves Binet's formula:

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} = \frac{\phi^n}{\sqrt{5}}(1 - (-q)^n). \tag{2.2}$$

Because $0 < q < 1$, the last factor is positive. Set

$$a = \log \phi, \qquad b = \log \sqrt{5}, \qquad \alpha = \frac{b}{a} = 1.672275938184554746\ldots. \tag{2.3}$$

Taking logarithms gives the exact identity

$$L_n := \log F_n = a(n - \alpha) + \varepsilon_n, \qquad \varepsilon_n = \log(1 - (-q)^n). \tag{2.4}$$

The slope a controls exponential growth. The intercept $-b$ becomes a horizontal shift α. The secondary characteristic root produces the alternating remainder. Keeping these three roles separate is useful: the intercept determines the rational bounds, while the secondary root determines the exponentially small oscillation.

Lemma 2.1. *For every $n \geq 1$, put $t_n = q^n/(1 - q^n)$. Then*

$$|\varepsilon_n| \leq t_n, \qquad \operatorname{sgn}(\varepsilon_n) = (-1)^{n+1}, \qquad \left|\varepsilon_n - (-1)^{n+1} q^n\right| \leq \frac{q^{2n}}{2(1-q^n)}. \tag{2.5}$$

Also $t_{n+h} \leq q^h t_n$ for $h \geq 0$.

Proof. For $|z| < 1$, the absolutely convergent power series

$$\log(1 - z) = -\sum_{k=1}^{\infty} \frac{z^k}{k}$$

implies $|\log(1-z)| \leq |z|/(1-|z|)$. Taking $z = (-q)^n$ proves the first bound. The logarithm is positive for odd n and negative for even n, which proves the sign assertion. Removing the first series term and using $1/k \leq 1/2$ for $k \geq 2$ proves the last estimate. Finally,

$$\frac{q^{n+h}}{1-q^{n+h}} \leq \frac{q^{n+h}}{1-q^n} = q^h t_n.$$

□

In particular, $L_n = an - b + O(q^n)$. The stronger expansion

$$\varepsilon_n = (-1)^{n+1} q^n - \frac{1}{2} q^{2n} + O(q^{3n}) \tag{2.6}$$

shows that the second-order correction has the same negative sign for both parities. Notice that small absolute errors in logarithms require careful handling when a conclusion concerns a floor or a strict inequality. The exact bound in Lemma 2.1, rather than the symbol $O(q^n)$ alone, will make those conclusions rigorous.

The elementary inequalities $\phi < \sqrt{5} < \phi^2$ give

$$1 < \alpha < 2. \tag{2.7}$$

For fixed n, the function

$$g_n(c) = \frac{n-c}{n+1-c} = 1 - \frac{1}{n+1-c} \tag{2.8}$$

has derivative $g_n'(c) = -(n+1-c)^{-2}$ wherever it is defined. Neglecting the remainder in (2.4) suggests $R_n \approx g_n(\alpha)$, and (2.7) then places this model between $g_n(2)$ and $g_n(1)$. This explains the form of the bounds. It is an explanation of their origin, not yet a proof of their uniform validity; that proof must control the remainder.

3 A uniform inequality for consecutive logarithms

Theorem 3.1. *For every integer $n \geq 3$,*

$$\frac{n-2}{n-1} < R_n < \frac{n-1}{n}. \tag{3.1}$$

Equivalently,

$$F_{n+1}^{n-2} < F_n^{n-1}, \qquad F_n^n < F_{n+1}^{n-1}. \tag{3.2}$$

Proof. Since $L_n, L_{n+1} > 0$, cross-multiplication shows that the two inequalities in (3.1) are respectively equivalent to positivity of

$$A_n = (n-1)L_n - (n-2)L_{n+1}, \tag{3.3}$$
$$B_n = (n-1)L_{n+1} - nL_n. \tag{3.4}$$

Substitution of (2.4) gives

$$A_n = 2a - b + (n-1)\varepsilon_n - (n-2)\varepsilon_{n+1}, \tag{3.5}$$
$$B_n = b - a + (n-1)\varepsilon_{n+1} - n\varepsilon_n. \tag{3.6}$$

The affine part has left a positive constant in each case. We now establish a common lower bound for these constants, entirely independently of decimal approximations.

Writing $s = \sqrt{5}$, we have

$$\frac{\phi^2}{\sqrt{5}} = \frac{3+s}{2s} > \frac{8}{7}, \qquad \frac{\sqrt{5}}{\phi} = \frac{5-s}{2} > \frac{8}{7}.$$

The first inequality follows from $21 > 9s$ and the second from $19 > 7s$; both follow by squaring positive quantities. Since $\log(1+x) > x/(1+x)$ for $x > 0$, it follows that

$$2a - b > \log(8/7) > \frac{1}{8}, \qquad b - a > \log(8/7) > \frac{1}{8}. \tag{3.7}$$

By Lemma 2.1, the absolute value of either remainder in (3.5)–(3.6) is at most

$$E_n = (2n-1)t_n = \frac{(2n-1)q^n}{1-q^n}.$$

The bound $q < 2/5$ yields, for $n \geq 5$,

$$E_n < \frac{(2n-1)(2/5)^n}{1-(2/5)^n} \leq \frac{9(2/5)^5}{1-(2/5)^5} = \frac{96}{1031} < \frac{1}{8}. \tag{3.8}$$

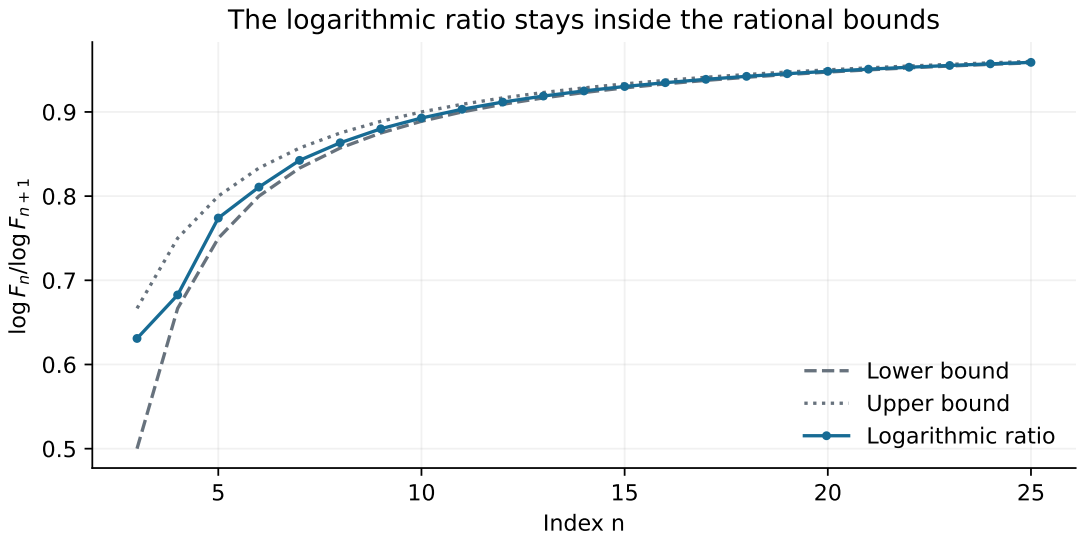


Figure 1: The logarithmic ratio and its rational enclosure for $3 \le n \le 25$. Points are computed from exact Fibonacci integers using 160-digit arithmetic; displayed positions are rounded only for plotting. The enclosure is proved in Theorem 3.1.

To justify the middle inequality, the ratio of successive terms of that upper bound is less than

$$\frac{2}{5}\frac{2n+1}{2n-1} \le \frac{2}{5}\frac{11}{9} < 1 \qquad (n \ge 5).$$

Thus the bound decreases. Equations (3.7) and (3.8) show $A_n, B_n > 0$ for $n \ge 5$.

The remaining cases use only $F_3 = 2$, $F_4 = 3$, $F_5 = 5$:

$$2^2 > 3, \quad 2^3 < 3^2; \qquad 3^3 > 5^2, \quad 3^4 < 5^3.$$

These are exactly (3.2) for $n = 3, 4$. Exponentiating (3.3)–(3.4) establishes the asserted equivalence and completes the proof. □

The proof illustrates a useful distinction between an asymptotic argument and a uniform one. Knowing merely that $\varepsilon_n \to 0$ would give the result for all sufficiently large n, but not identify an onset. The rational estimate (3.8) reduces the entire infinite assertion to two finite checks. It also avoids basing positivity on rounded values of a and b.

The interval in (3.1) has exact width

$$\frac{n-1}{n} - \frac{n-2}{n-1} = \frac{1}{n(n-1)}. \tag{3.9}$$

Its width is therefore of order n^{-2}. Both endpoints approach 1 at order n^{-1}, but their separation is one order smaller. This is why the bound contains more information than the simple limit $R_n \to 1$.

4 Monotonicity, concavity, and root sequences

The strength of the enclosure is especially clear when the index is increased by one. The upper endpoint of the interval for R_n equals the lower endpoint of the interval for R_{n+1}. Thus the intervals are ordered without overlapping interiors.

Corollary 4.1. *The sequence $(R_n)_{n\geq 3}$ is strictly increasing and converges to 1. More precisely,*

$$R_n < \frac{n-1}{n} < R_{n+1} \qquad (n \geq 3). \tag{4.1}$$

Proof. The first inequality is the upper bound in Theorem 3.1. Applying the lower bound at $n+1$ gives the second. The limit follows by squeezing between the two endpoints of (3.1). □

Corollary 4.2. *The positive sequence $(L_n)_{n\geq 3} = (\log F_n)_{n\geq 3}$ is strictly log-concave:*

$$L_{n+1}^2 > L_n L_{n+2} \qquad (n \geq 3). \tag{4.2}$$

Equivalently, $(\log\log F_n)_{n\geq 3}$ is a strictly concave sequence.

Proof. The inequality $R_n < R_{n+1}$ is $L_n/L_{n+1} < L_{n+1}/L_{n+2}$. All factors are positive, so it is equivalent to (4.2). Taking logarithms gives the concavity statement. □

It is necessary to distinguish this assertion from log-concavity of the Fibonacci numbers themselves. Cassini's identity is

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n. \tag{4.3}$$

It follows, for example, by taking determinants of powers of the recurrence matrix $\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$, whose determinant is -1. Identity (4.3) shows that the original sequence alternates between log-convexity at even middle indices and log-concavity at odd ones. By contrast, after one logarithm the resulting positive sequence has uniform strict log-concavity from index 3 onward. Taking a logarithm has not simply preserved the original curvature pattern.

Corollary 4.3. *For $n \geq 3$, the sequences*

$$U_n = F_n^{1/(n-1)}, \qquad V_n = F_n^{1/(n-2)}$$

are respectively strictly increasing and strictly decreasing, and both converge to ϕ. Consequently

$$U_n < \phi < V_n \qquad (n \geq 3). \tag{4.4}$$

Also $(F_n^{1/n})_{n\geq 2}$ is strictly increasing to ϕ.

Proof. The upper bound in (3.1) is equivalent to

$$\frac{L_n}{n-1} < \frac{L_{n+1}}{n},$$

which proves monotonicity of U_n. The lower bound is equivalent to

$$\frac{L_n}{n-2} > \frac{L_{n+1}}{n-1},$$

which proves monotonicity of V_n. From (2.4), both normalized logarithms approach a, giving the stated limits. Strict monotonicity toward a finite limit implies (4.4). Finally $(n-1)/n < n/(n+1)$, so Theorem 3.1 implies $L_n/n < L_{n+1}/(n+1)$ for $n \geq 3$; the transition from $n = 2$ to 3 is immediate. □

The last corollary gives two monotone approximations to the same exponential growth rate. It differs from the stronger root-quotient monotonicity theorem of Hou, Sun, and Wen [3]: that theorem controls successive increments of L_n/n. Our argument controls two shifted root sequences and the ratio of successive logarithms. These properties are related, but they should not be treated as interchangeable.

5 Effective shifts and the optimal uniform enclosure

The constants 1 and 2 provide a convenient integer enclosure, but they need not be the best possible real constants. To formulate that question precisely, we absorb the exact ratio into a shift parameter.

Definition 5.1. For $n \geq 3$, define

$$\delta_n = n - \frac{L_n}{L_{n+1} - L_n}. \tag{5.1}$$

This is the effective shift at index n.

Since $F_{n+1} > F_n > 1$, the denominator is positive. A direct calculation gives

$$R_n = g_n(\delta_n). \tag{5.2}$$

Thus (3.1) is exactly the assertion $1 < \delta_n < 2$. The effective shift also has an interpretation in terms of discrete growth: $L_{n+1} - L_n$ is the local slope, while L_n is the current height. The line with that local slope would reach height zero at index δ_n.

Put $d_n = \varepsilon_{n+1} - \varepsilon_n$. Substituting (2.4) into (5.1) yields

$$\delta_n - \alpha = \frac{(n - \alpha)d_n - \varepsilon_n}{a + d_n}. \tag{5.3}$$

The denominator equals $L_{n+1} - L_n$ and is positive. Moreover d_n has sign $(-1)^n$, because consecutive remainders have opposite signs. Since $n - \alpha > 0$,

$$\delta_n < \alpha \quad \text{for odd } n, \qquad \delta_n > \alpha \quad \text{for even } n. \tag{5.4}$$

No numerical experiment is required to establish this alternation. It records the negative secondary characteristic root.

Whenever $a > (1 + q)t_n$, Lemma 2.1 gives the quantitative estimate

$$|\delta_n - \alpha| \leq \frac{((n - \alpha)(1 + q) + 1)t_n}{a - (1 + q)t_n}. \tag{5.5}$$

In particular, $\delta_n = \alpha + O(nq^n)$. This identifies the unique limiting shift even though the individual shifts alternate around it.

Theorem 5.2. *The global minimum and maximum of $(\delta_n)_{n\geq 3}$ are*

$$\delta_- = \delta_3 = 3 - \frac{\log 2}{\log(3/2)} = 1.290488708648545223\ldots, \tag{5.6}$$

$$\delta_+ = \delta_4 = 4 - \frac{\log 3}{\log(5/3)} = 1.849339896912876491\ldots. \tag{5.7}$$

Consequently the optimal enclosure by real constant shifts with positive denominators on the whole index domain is

$$g_n(\delta_+) \leq R_n \leq g_n(\delta_-) \qquad (n \geq 3). \tag{5.8}$$

Equality occurs only at $n = 4$ in the lower bound and at $n = 3$ in the upper bound. No smaller lower-bound shift or larger upper-bound shift works on the whole domain.

Proof. We first give a uniform tail estimate. Elementary bounds, justified in Appendix A, are

$$q < \frac{2}{5}, \qquad a > \frac{4}{9}, \qquad \alpha > \frac{3}{2}.$$

For $n \geq 6$, let $s_n = (2/5)^n/(1-(2/5)^n)$. The numerator in (5.5) is less than

$$\left(\frac{7n}{5} - \frac{11}{10}\right) s_n,$$

while its denominator is greater than $4/9 - (7/5)s_n$. The displayed numerator bound decreases for $n \geq 6$, as is seen by taking the ratio of its successive terms. The denominator bound increases. Since $s_6 = 64/15561$, we obtain

$$|\delta_n - \alpha| < \frac{(73/10)(64/15561)}{4/9 - (7/5)(64/15561)} = \frac{584}{8533} < \frac{1}{12} \qquad (n \geq 6). \tag{5.9}$$

The following intentionally coarse rational enclosures are certified by the logarithm series in Appendix A:

$$\begin{aligned} 1.67 < \alpha < 1.68, \qquad & 1.29 < \delta_3 < 1.30, \\ 1.84 < \delta_4 < 1.85, \qquad & 1.57 < \delta_5 < 1.58. \end{aligned} \tag{5.10}$$

Combining (5.9) with (5.10) places every δ_n, $n \geq 6$, strictly between $1.67-1/12$ and $1.68+1/12$. This interval lies strictly inside $(1.30, 1.84)$. The case $n = 5$ also lies strictly between δ_3 and δ_4. Hence those two indices give the unique extrema.

The function g_n is strictly decreasing on the relevant interval. Applying it to $\delta_- \leq \delta_n \leq \delta_+$ proves (5.8) and identifies equality. At $n = 4$, a smaller lower-bound shift would give a value larger than R_4. At $n = 3$, a larger upper-bound shift would give a value smaller than R_3. This proves optimality. □

The optimal constants are determined by small indices, whereas the limiting constant is α. This is a useful example of the difference between uniform and asymptotic optimization. The approximation $g_n(\alpha)$ is exponentially accurate for large n, but it lies above the true ratio at even indices and below it at odd indices. It cannot itself be a one-sided bound for all indices.

6 Exact remainders and asymptotic expansions

Proposition 6.1. *For every $n \geq 3$,*

$$R_n - g_n(\alpha) = \frac{(n+1-\alpha)\varepsilon_n - (n-\alpha)\varepsilon_{n+1}}{(n+1-\alpha)\big(a(n+1-\alpha) + \varepsilon_{n+1}\big)}. \tag{6.1}$$

The error is positive for odd n and negative for even n. Its absolute value is bounded by

$$\frac{\big((n+1-\alpha) + (n-\alpha)q\big)t_n}{(n+1-\alpha)\big(a(n+1-\alpha) - t_{n+1}\big)}. \tag{6.2}$$

The denominator in (6.2) is positive for $n \geq 3$.

Proof. Subtract $(n-\alpha)/(n+1-\alpha)$ from L_n/L_{n+1} and use (2.4). This gives (6.1). Its denominator is positive. The numerator has the sign of ε_n, since its two summands have that same sign. Lemma 2.1 proves the bound. For positivity of its denominator, $n+1-\alpha > 2$ when $n \geq 3$, while $a > 4/9$ and $t_{n+1} < t_4 < 16/609$. Thus $a(n+1-\alpha) > 8/9 > t_{n+1}$. □

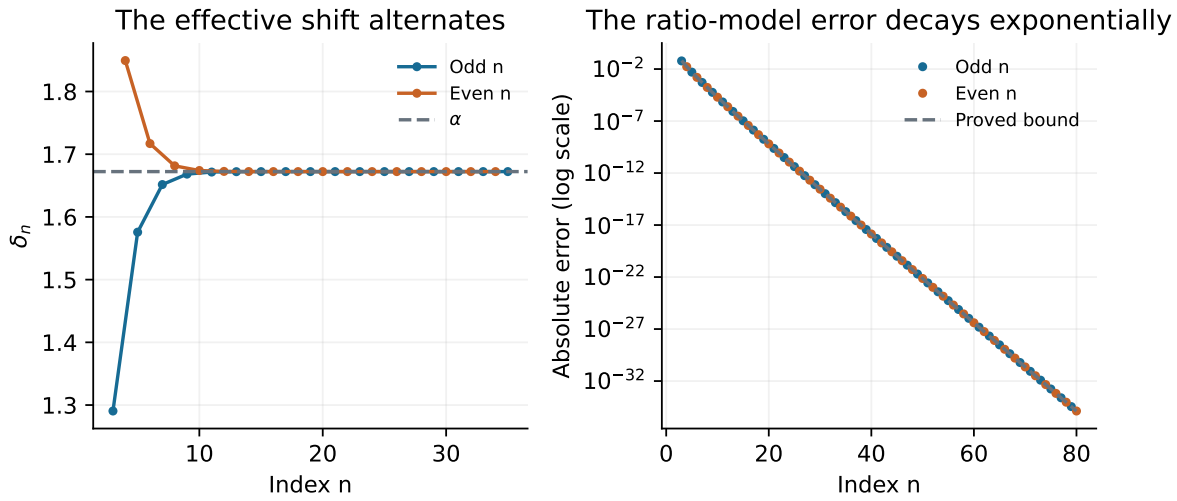


Figure 2: Left: the effective shift lies below α at odd indices and above it at even indices. Right: the absolute error of the affine ratio model decreases exponentially; the dashed curve is the explicit bound in Proposition 6.1. The vertical scale on the right is logarithmic.

The remainder is $O(q^n/n)$, smaller than every fixed algebraic power of $1/n$. Therefore a rational model captures all orders of the algebraic asymptotic expansion. More explicitly, Lemma 2.1 and (6.1) give

$$R_n - g_n(\alpha) = \frac{(-1)^{n+1}(1+q)q^n}{an} + O\left(\frac{q^n}{n^2} + \frac{q^{2n}}{n}\right). \tag{6.3}$$

Here the leading coefficient comes from the combination of the remainder at n with the oppositely signed remainder at $n+1$. A statement that merely says $R_n - g_n(\alpha) = O(q^n/n)$ would conceal this parity information.

Expanding the rational model at infinity gives the next theorem.

Theorem 6.2. *As $n \to \infty$,*

$$R_n = 1 - \frac{1}{n} + \frac{1-\alpha}{n^2} - \frac{(1-\alpha)^2}{n^3} + O(n^{-4}). \tag{6.4}$$

In particular,

$$n(1-R_n) \longrightarrow 1, \qquad n^2\left(R_n - 1 + \frac{1}{n}\right) \longrightarrow 1-\alpha. \tag{6.5}$$

Proof. Write $d = 1-\alpha$. Then

$$g_n(\alpha) = 1 - \frac{1}{n+d} = 1 - \frac{1}{n} + \frac{d}{n^2} - \frac{d^2}{n^3} + O(n^{-4}).$$

Proposition 6.1 shows that adding the exact remainder does not alter any of these coefficients. □

Another useful statistic records the ratio's position inside its proved interval. Define $\ell_n = (n-2)/(n-1)$, $u_n = (n-1)/n$, and

$$P_n = \frac{R_n - \ell_n}{u_n - \ell_n}.$$

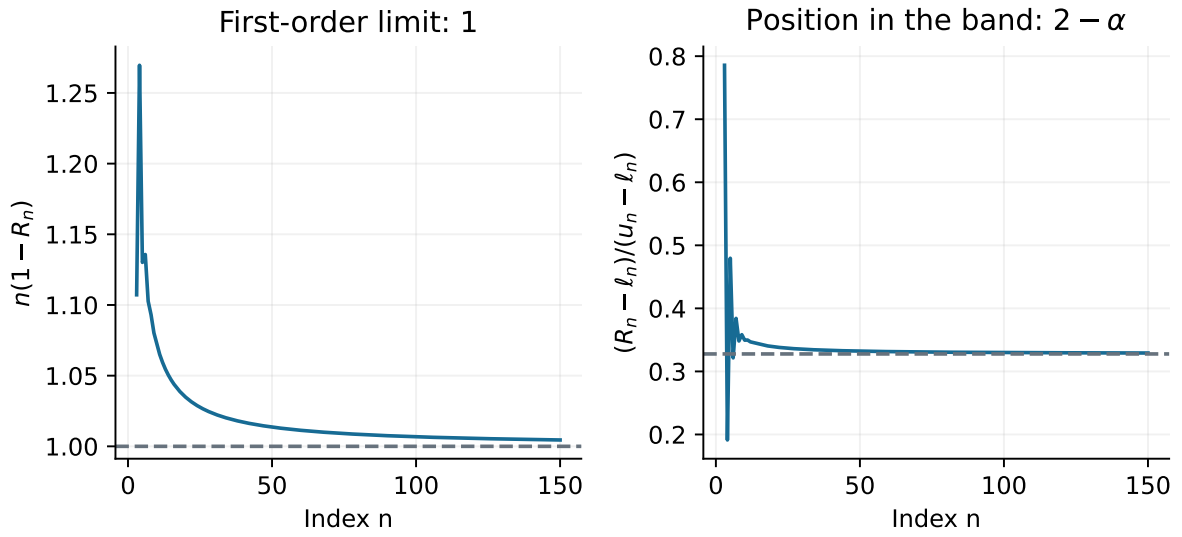


Figure 3: Two normalized limits for $3 \le n \le 150$. Left: $n(1-R_n)$ approaches 1. Right: the position inside the rational band approaches $2-\alpha$. The early oscillation in the right panel reflects the alternating Binet remainder.

Using (3.9) and (6.4),

$$P_n \longrightarrow 2-\alpha = 0.327724061815445254\ldots. \tag{6.6}$$

The true ratio therefore approaches a point about one third of the way from the lower endpoint to the upper endpoint. It does not asymptotically sit at the midpoint. The exact shift and the algebraic width together explain that location.

7 Ratios with nonadjacent indices

The consecutive-index theorem extends immediately to a gap of any positive integer length. For $n \ge 3$ and $h \ge 1$, set

$$R_{n,h} = \frac{L_n}{L_{n+h}}.$$

Theorem 7.1. *For every $n \ge 3$ and $h \ge 1$,*

$$\frac{n-2}{n+h-2} < R_{n,h} < \frac{n-1}{n+h-1}. \tag{7.1}$$

For each fixed h, $R_{n,h}$ increases strictly with n and approaches 1.

Proof. The positive quotients telescope:

$$R_{n,h} = \prod_{j=0}^{h-1} R_{n+j}.$$

Multiplying the lower bounds in Theorem 3.1 gives

$$\prod_{j=0}^{h-1} \frac{n+j-2}{n+j-1} = \frac{n-2}{n+h-2};$$

multiplying the upper bounds gives the other endpoint. Each factor in the product strictly increases when n increases, proving monotonicity. For fixed h, both endpoints approach 1. □

The width of this enclosure is

$$\frac{h}{(n+h-2)(n+h-1)}. \tag{7.2}$$

For fixed h it is again of order n^{-2}. The proof is simpler than repeating all estimates, but Binet's formula adds the exact error and its sign:

$$R_{n,h} - \frac{n-\alpha}{n+h-\alpha} = \frac{(n+h-\alpha)\varepsilon_n - (n-\alpha)\varepsilon_{n+h}}{(n+h-\alpha)L_{n+h}}. \tag{7.3}$$

If h is odd, the two remainders have opposite signs. If h is even, they have the same sign, but $|\varepsilon_{n+h}| < |\varepsilon_n|$ and the first coefficient in the numerator is larger. Thus the numerator always has the sign of ε_n. The alternating placement relative to the affine model persists for every gap length.

For each fixed h, (7.3) gives

$$R_{n,h} = 1 - \frac{h}{n} + \frac{h(h-\alpha)}{n^2} + O(n^{-3}), \qquad n(1-R_{n,h}) \longrightarrow h. \tag{7.4}$$

The phrase "fixed h" is important: this expansion is not uniform for gaps growing arbitrarily fast with n. If $h = h_n$ and $h_n/n \to \lambda \geq 0$, then the exact logarithmic model instead yields

$$R_{n,h_n} \longrightarrow \frac{1}{1+\lambda}. \tag{7.5}$$

The rational enclosure proves the same conclusion directly. If $h_n/n \to \infty$, the ratio approaches 0. These regimes distinguish comparisons over a bounded number of steps from comparisons over a macroscopic part of the index scale.

8 A dominant-exponential principle

The Fibonacci calculation depends on a general feature of recurrence sequences: one exponential term dominates, and the other terms decay relative to it. We now state that principle without requiring integer terms or a particular recurrence.

Theorem 8.1. *Suppose that, for all sufficiently large n,*

$$u_n = C\rho^n(1+\eta_n), \qquad C > 0, \quad \rho > 1, \quad |\eta_n| \leq M\tau^n, \quad 0 < \tau < 1, \tag{8.1}$$

where M is a fixed positive constant and $1+\eta_n > 0$. Put

$$A = \log\rho, \qquad \beta = -\frac{\log C}{A}.$$

For any fixed real numbers $c_- < \beta < c_+$, one has, for all sufficiently large n,

$$\frac{n-c_+}{n+1-c_+} < \frac{\log u_n}{\log u_{n+1}} < \frac{n-c_-}{n+1-c_-}. \tag{8.2}$$

Moreover,

$$\frac{\log u_n}{\log u_{n+1}} = 1 - \frac{1}{n} + \frac{1-\beta}{n^2} + O(n^{-3}), \tag{8.3}$$

and these logarithmic ratios are eventually strictly increasing.

Proof. Write $e_n = \log(1+\eta_n)$. For large n, $M\tau^n < 1$ and

$$|e_n| \le \frac{M\tau^n}{1-M\tau^n}.$$

Thus $\log u_n = A(n-\beta) + e_n$, with $e_n = O(\tau^n)$. Both $\log u_n$ and its increment are positive eventually. For any fixed c, cross-multiplication gives

$$\begin{aligned}(n+1-c)\log u_n - (n-c)\log u_{n+1} \\ = A(c-\beta) + (n+1-c)e_n - (n-c)e_{n+1}.\end{aligned} \tag{8.4}$$

The last two terms are $O(n\tau^n)$ and tend to zero. Choosing $c = c_+$ gives eventual positivity, while choosing $c = c_-$ gives eventual negativity, proving (8.2). An exact subtraction as in Proposition 6.1 shows that the quotient differs from $(n-\beta)/(n+1-\beta)$ by $O(\tau^n/n)$. Expanding this model proves (8.3).

For monotonicity, set $K_n = \log u_n$. Direct expansion gives

$$K_{n+1}^2 - K_n K_{n+2} = A^2 + O(n\tau^n).$$

The positive constant A^2 dominates the remainder eventually. Hence $K_n/K_{n+1} < K_{n+1}/K_{n+2}$ for large n. □

The theorem gives an asymptotic enclosure; it does not supply a universal starting index for arbitrary C, M, τ. A concrete sequence requires a margin estimate and a finite verification, as in Section 3. The prefactor C is essential. The dominant root controls the first-order term $-1/n$, but C controls the next coefficient through β.

8.1 Lucas and Pell examples

Let $\mathcal{L}_n = \phi^n + \psi^n$ be the Lucas sequence, with $\mathcal{L}_0 = 2$, $\mathcal{L}_1 = 1$. Then

$$\mathcal{L}_n = \phi^n\bigl(1 + (-q)^n\bigr).$$

Here $C = 1$ and $\beta = 0$. Its logarithmic ratio approaches $n/(n+1)$ with an exponentially small error, but the error is positive at even indices and negative at odd indices. This follows by applying the exact subtraction formula to $e_n = \log(1+(-q)^n)$. An integer limiting shift therefore need not give a one-sided bound: the secondary root can put the ratio on both sides infinitely often.

For the Pell sequence $P_0 = 0$, $P_1 = 1$, $P_{n+2} = 2P_{n+1} + P_n$, the characteristic roots are $1+\sqrt{2}$ and $1-\sqrt{2}$. Its prefactor and limiting shift are

$$C = \frac{1}{2\sqrt{2}}, \qquad \rho = 1+\sqrt{2}, \qquad \beta = \frac{\log(2\sqrt{2})}{\log(1+\sqrt{2})} = 1.179659552036092327\ldots.$$

Since $1 < \beta < 2$, the same integer shifts as for Fibonacci give eventual bounds. We assert only eventual validity here; Theorem 8.1 alone does not establish validity at every small Pell index.

8.2 Changes of scale and limitations

Multiplying a sequence by a positive constant c changes its shift to

$$\beta' = \beta - \frac{\log c}{\log \rho}.$$

Thus a common logarithmic base does not change the ratio, but scaling the original terms generally does. Similarly, replacing u_n by u_{n+s} changes the limiting shift from β to $\beta - s$. This accounts for the Fibonacci indexing issue in the introduction.

A simple positive dominant characteristic root with all other roots of smaller modulus produces (8.1), provided its coefficient is positive. A repeated dominant root can instead introduce a polynomial factor. For example, $u_n = n^k \rho^n$ has

$$\log u_n = n \log \rho + k \log n.$$

Its effective shift is $-(k/\log \rho) \log n + O(1)$, rather than a constant. The leading limit $n(1 - R_n) \to 1$ survives, but constant-shift enclosures need not. The absence of a polynomial factor is therefore a substantive assumption of the theorem.

9 Digit counts and Benford's law

Logarithms describe more than growth rates. Their integer parts count digits, while their fractional parts determine the leading significand. These operations have different sensitivities to the remainder in (2.4).

9.1 Digit counts and safe rounding

For a positive integer m, its number of base-B digits, $B \geq 2$ an integer, is

$$D_B(m) = \lfloor \log_B m \rfloor + 1.$$

Consequently

$$D_B(F_n) = \left\lfloor n \log_B \phi - \log_B \sqrt{5} + \frac{\varepsilon_n}{\log B} \right\rfloor + 1. \tag{9.1}$$

The leading-order growth rate is $n \log_B \phi$. In base ten, $\log_{10} \phi = 0.208987640249978734\ldots$, so roughly five indices add one decimal digit on average. The term "on average" matters: digit-count increments follow the fractional rotation of the logarithm rather than a fixed period.

It is tempting to omit ε_n before applying the floor. An exponentially small correction is not automatically harmless near an integer. A sufficient certificate for omitting it is

$$\operatorname{dist}\left(n \log_B \phi - \log_B \sqrt{5}, \mathbb{Z}\right) > \frac{t_n}{\log B}. \tag{9.2}$$

Under this condition, adding the remainder cannot cross a floor boundary. Without it, one should evaluate a higher-precision interval or use exact integer arithmetic. For example, the exact decimal digit counts of F_{100}, F_{1000}, and F_{10000} are 21, 209, and 2090. These values were obtained from integer representations, not by assuming that the uncorrected floor formula always works.

The identity also reveals a small-index pitfall: $F_2 = 1$ has logarithm zero, while the affine logarithmic model gives $2a - b > 0$. Even though the digit count happens to agree in that instance, the correction is large enough relative to the distance from a boundary to require attention. The word "exact" should refer to (9.1), not to a floor taken after discarding its remainder.

9.2 Irrationality of the logarithmic slope

The leading-digit result depends on the irrationality of $\log_B \phi$. It is not enough that ϕ itself is irrational: an irrational number can have a rational logarithm in a given base.

Lemma 9.1. *For every integer base $B \geq 2$, $\log_B \phi$ is irrational.*

Proof. If $\log_B \phi = p/r$ with positive integers p, r, then $\phi^r = B^p$. Applying the nontrivial field automorphism of $\mathbb{Q}(\sqrt{5})$ sends ϕ to ψ and fixes B^p, giving $\psi^r = B^p$. But $|\psi|^r < 1$ and $B^p > 1$, a contradiction. □

9.3 Uniform distribution modulo one

A sequence (x_n) is uniformly distributed modulo one if, for every interval $[u, v) \subseteq [0, 1)$,

$$\lim_{N\to\infty} \frac{1}{N} \#\{1 \leq n \leq N : \{x_n\} \in [u, v)\} = v - u.$$

Weyl's criterion [5] characterizes this property by vanishing averages of $e^{2\pi i k x_n}$ for every nonzero integer k. For an irrational θ and a fixed γ, the geometric sum gives

$$\left| \frac{1}{N} \sum_{n=1}^{N} e^{2\pi i k(n\theta+\gamma)} \right| \leq \frac{2}{N|1 - e^{2\pi i k\theta}|} \longrightarrow 0. \tag{9.3}$$

The denominator is nonzero because θ is irrational. Hence $(n\theta + \gamma)$ is uniformly distributed modulo one.

This remains true if a perturbation tending to zero is added. Indeed, for $r_n \to 0$,

$$\left| e^{2\pi i k(x_n+r_n)} - e^{2\pi i k x_n} \right| \leq 2\pi |k|, |r_n|,$$

and the average of the right-hand side tends to zero by the elementary Cesaro principle. This proves stability through Weyl's criterion. It also explains why an error that can matter at a particular digit boundary does not alter the limiting digit distribution.

Taking

$$\theta = \log_B \phi, \qquad \gamma = -\log_B \sqrt{5}, \qquad r_n = \varepsilon_n / \log B,$$

Lemma 9.1 and (2.4) show that $(\log_B F_n)$ is uniformly distributed modulo one. This conclusion concerns the positive terms $n \geq 1$; deleting any finite initial segment does not affect the limit.

Theorem 9.2 (Fibonacci leading-digit law)**.** *For every integer $B \geq 2$ and each digit $d \in \{1, \ldots, B-1\}$,*

$$\lim_{N\to\infty} \frac{1}{N} \#\{1 \leq n \leq N : F_n \text{ begins with } d \text{ in base } B\} = \log_B \left(1 + \frac{1}{d}\right). \tag{9.4}$$

Proof. A positive number x has leading base-B digit d if and only if

$$\{\log_B x\} \in [\log_B d, \log_B(d+1)).$$

Uniform distribution assigns this interval its length, namely $\log_B(d+1) - \log_B d$. Applying the preceding argument to Fibonacci logarithms proves (9.4). □

This is the classical Fibonacci Benford theorem, established in the literature cited above, particularly Washington [2]. The proof here emphasizes its connection with the same remainder that controls our ratio inequality. In base ten the limiting frequencies for leading digits 1, 2, and 9 are approximately 30.1030%, 17.6091%, and 4.5757%. The probabilities add to one because their logarithms telescope.

The method gives a multi-digit extension. For a fixed positive integer s with k base-B digits, the limiting proportion of Fibonacci numbers beginning with the block represented by s is

$$\log_B\left(1+\frac{1}{s}\right). \tag{9.5}$$

For example, the decimal prefix 12 corresponds to the half-open interval with endpoints $\log_{10}(1.2)$ and $\log_{10}(1.3)$. A finite number of initial terms with too few digits to have that prefix does not affect its limiting frequency.

9.4 Global frequencies do not imply local independence

The preceding theorem describes a single leading digit sampled along the sequence. It does not imply that two consecutive leading digits behave like independent samples. This distinction is emphasized in the local Benford literature [7]. It can be demonstrated directly here, without importing a more general local-distribution theorem.

In base ten, put $\theta = \log_{10}\phi$, and let

$$I_d = [\log_{10} d, \log_{10}(d+1)).$$

The fractional logarithms of consecutive terms are asymptotically related by a rotation through θ. Thus the limiting frequency of the leading-digit pair (d, e) is the circle length

$$|I_d \cap (I_e - \theta)|, \tag{9.6}$$

where translation is interpreted modulo one. To justify this, the unperturbed phases are uniformly distributed, and the perturbations vanish. Membership can change only in shrinking neighborhoods of the finitely many interval endpoints; those neighborhoods have limiting frequency tending to zero.

Let $d_{10}(x)$ denote the leading decimal digit of a positive number x. Consider specifically $(d, e) = (1, 2)$. The translated interval

$$I_2 - \theta = [\log_{10}(2/\phi), \log_{10}(3/\phi))$$

lies wholly inside $I_1 = [0, \log_{10} 2)$. The lower endpoint is positive because $\phi < 2$, and the upper endpoint is less than $\log_{10} 2$ because $\phi > 3/2$. Therefore

$$\begin{aligned}\lim_{N\to\infty}\frac{1}{N-1}\#\{1 \le n < N : d_{10}(F_n) = 1,\ d_{10}(F_{n+1}) = 2\} &= \log_{10}(3/2)\\ &= 0.1760912590\ldots.\end{aligned} \tag{9.7}$$

If the digits were independent with Benford marginals, the limit would instead be

$$\log_{10} 2\,\log_{10}(3/2) = 0.0530087509\ldots. \tag{9.8}$$

The difference is substantial. An irrational rotation produces correct one-dimensional frequencies while retaining a rigid relation between successive phases. The same deterministic structure underlies both the success of Benford's law and the failure of independence.

n	R_n	δ_n	$n(1-R_n)$	P_n
3	0.630929754	1.290488709	1.107210739	0.785578521
4	0.682606194	1.849339897	1.269575222	0.191274334
5	0.773976032	1.575690458	1.130119842	0.479520633
10	0.892772962	1.673994035	1.072270384	0.349566542
20	0.948260850	1.672276177	1.034782998	0.339123040
50	0.979727425	1.672275938	1.013628765	0.332190536
100	0.989932317	1.672275938	1.006768261	0.329942184

Table 1: Selected logarithmic statistics, rounded to nine decimal places. The effective shift approaches α exponentially up to a factor of n, whereas the normalized band position approaches $2-\alpha$ at an algebraic rate. The differing rates explain why the shift appears stable before the last column does.

10 Reproducible numerical investigation

The computations serve three purposes: checking formulas at finite indices, displaying scales that are difficult to see in unnormalized quantities, and distinguishing a global frequency law from local digit dependence. None of the finite checks replaces the proofs of the infinite assertions.

10.1 Arithmetic and verification design

Fibonacci values are generated by the integer recurrence (1.1). All logarithmic computations use `mpmath` with 160 decimal digits of working precision. Plot coordinates are converted to ordinary floating-point values only after the high-precision calculations. For tiny approximation errors, the program evaluates (6.1) directly, avoiding subtraction of two nearly equal plotted ratios.

The program verifies the strict enclosure and monotonicity for every index $3 \le n \le 5000$, a total of 4998 indices. It also checks (7.1) for $h = 2, 5, 10$ throughout that range. For $3 \le n \le 150$, it compares the direct quotient with the exact remainder formula, verifies its alternating sign, and checks the explicit error bound. The smallest computed distance from either endpoint of (3.1) in the 4998-index run is about 1.3113×10^{-8}. This value describes the finite experiment, not a lower bound uniform in n; the margins approach zero as the index grows.

The constants used in Theorem 5.2 have an additional exact-arithmetic check. The program uses rational enclosures of $\sqrt{5}$ and finite logarithm series with explicit rational tails to certify (5.10). This calculation does not depend on a rounded decimal approximation. An independent interval-arithmetic calculation is recorded as well. The algebraic tail bound $584/8533 < 1/12$ is checked using rational arithmetic.

Table 1 highlights the special role of the smallest indices. The first two shifts are the global extrema. By index 20, the effective shift is already close to its limit at the seventh decimal place, while $n(1-R_n)$ remains visibly above its limiting value. These are not conflicting convergence patterns: the shift correction is of order nq^n, while the latter statistic has an algebraic correction of order $1/n$.

The error panel of Figure 2 uses the analytic remainder. Subtracting two double-precision ratios would soon replace the true error with rounding noise. Direct evaluation of the explicit formula prevents that loss of precision.

Digit	$N = 100$	$N = 1000$	$N = 10000$	Observed frequency	Benford limit
1	30	301	3011	0.3011	0.301030
2	18	177	1762	0.1762	0.176091
3	13	125	1250	0.1250	0.124939
4	9	96	968	0.0968	0.096910
5	8	80	792	0.0792	0.079181
6	6	67	668	0.0668	0.066947
7	5	56	580	0.0580	0.057992
8	7	53	513	0.0513	0.051153
9	4	45	456	0.0456	0.045757

Table 2: Exact first-digit counts and limiting frequencies. Every count column sums to its sample size. The empirical sample is deterministic; the limiting proportions are supplied by Theorem 9.2.

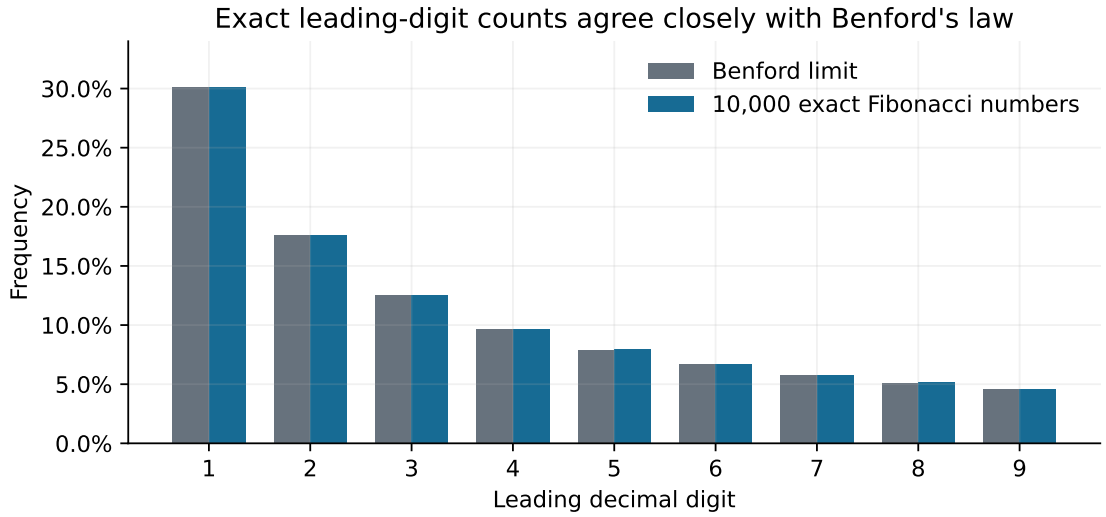


Figure 4: Leading-digit frequencies among $F_1, \ldots, F_{10000}$, computed from exact integers, compared with the Benford limits. Bar heights are measured from zero. The close marginal fit does not imply independence between neighboring digits.

10.2 Leading-digit experiments

Leading decimal digits are obtained directly from exact integer representations of F_n, for $1 \leq n \leq 10000$. The initial repeated value $F_1 = F_2 = 1$ is counted twice, because the sample is a sequence of indices. This convention is relevant to small samples and negligible for limiting densities. Counts at $N = 100$, 1000, and 10000 are shown in Table 2.

For $N = 10000$, the largest absolute difference between an empirical proportion and its Benford limit is approximately 0.0001575, or 0.01575 percentage points. Figure 4 displays these proportions. A close fit at a finite sample size illustrates the theorem but does not prove it or provide a universal convergence rate.

Among the 9999 consecutive pairs (F_n, F_{n+1}) with $1 \leq n < 10000$, exactly 1762 have leading digits $(1, 2)$. The observed frequency is therefore

$$\frac{1762}{9999} = 0.1762176217\ldots.$$

It agrees closely with the rotation-based limit $0.1760912590\ldots$ in (9.7), and is far from the in-

dependence prediction 0.0530087509 . . . . No independent-sampling significance test is attached to this comparison: the sample consists of a deterministic recurrence with known dependence. The informative comparison is between two mathematical models of the same limiting statistic.

The source bundle contains the exact-count data, high-precision ratio data, verification metadata, the plotting program, and the LaTeX manuscript. The program's assertions detect a failed verification by stopping execution. Successful completion is a finite computational result; the theorem statements remain supported by their proofs.

11 Discussion and further directions

The analysis is built on a single decomposition,

$$\log F_n = n \log \phi - \log \sqrt{5} + \log(1 - (-\phi^{-2})^n).$$

Its three components govern different phenomena. The slope determines the first-order growth of logarithms and the limiting value $n(1-R_n) \to 1$. The intercept determines the constant shift, the second-order coefficient, and the ratio's position in its rational band. The exponentially small remainder produces parity and controls whether a point lies above or below the affine model. Treating all three components as merely "approximately linear" would lose precisely the information needed for strict uniform inequalities.

The best integer bounds and the best real-shift bounds answer different questions. The integer enclosure is simple, has touching endpoints at consecutive indices, and proves monotonicity immediately. The optimal real-shift enclosure is tighter and characterizes the influence of initial terms. The asymptotic model is much more accurate at large indices, but it is two-sided across parity. There is no single approximation that automatically serves all these purposes.

The digit analysis also makes a conceptual distinction. A pointwise error can influence a digit boundary at a particular index, yet a perturbation tending to zero does not change an equidistribution limit. Conversely, a globally equidistributed sequence can have strong local dependence. These facts explain why the remainder must be bounded carefully for safe floor calculations, while the limiting Benford law follows from the irrational slope and stability of uniform distribution.

11.1 A connection with Diophantine equations

Suppose that a Fibonacci number is a perfect power, $F_n = y^m$ with integers $y \geq 2$ and $m \geq 2$. Equation (2.4) then becomes

$$\Lambda = n \log \phi - m \log y - \log \sqrt{5} = -\varepsilon_n, \qquad 0 < |\Lambda| \leq t_n. \tag{11.1}$$

The strict positivity follows because $q^n \neq 0$ and hence $\varepsilon_n \neq 0$. Thus an integer equation forces a nonzero linear combination of logarithms to be exponentially small. Lower bounds for such expressions can restrict the possible indices, but the upper bound alone does not solve the equation. The classification proved by Bugeaud, Mignotte, and Siksek [8] requires much deeper tools; the present identity explains only the elementary entry point to that approach.

11.2 Projects suggested by the analysis

Several further investigations follow naturally. One can seek a direct proof describing variation of δ_n within each parity class, rather than only locating its global extrema. One can develop

explicit onset bounds in Theorem 8.1 as functions of the spectral gap and prefactor. One can also quantify digit discrepancy using rational approximations to $\log_B \phi$, since irrationality alone gives convergence without a useful uniform numerical rate. These are proposed directions for study; we do not assert that they are unresolved in the wider literature.

For recurrences with several subdominant roots, the correction need not alternate simply. Complex roots can produce several oscillatory frequencies. The dominant-exponential theorem still gives eventual constant-shift bounds if the relative error decays exponentially, but obtaining optimal constants over every index requires more detailed information about the oscillation. Repeated dominant roots present a different issue, because polynomial factors cause the effective shift to drift logarithmically.

12 Conclusion

Binet's formula turns the motivating inequality into a controlled comparison between an affine logarithmic model and an exponentially small correction. The resulting uniform theorem proves more than convergence: it orders consecutive logarithmic ratios, yields strict log-concavity, and gives monotone root approximations. The effective shift makes the role of the intercept exact and identifies the best real constant shifts on the full domain. Nonadjacent ratios and dominant-exponential sequences follow the same mechanism, while fractional logarithms lead to digit counts, Benford's law, and an explicit failure of local independence. Together, these results show how one closed form supports several distinct kinds of logarithmic analysis, provided that approximation, indexing, and finite-index control are kept precise.

A Exact certification of constants

This appendix supplies explicit certificates for the numerical-looking inequalities used in Theorem 5.2. Their proof does not require treating a rounded decimal as exact.

First, $q < 2/5$ follows from $\sqrt{5} > 11/5$. Also $\alpha > 3/2$ follows from $5 > \phi^3 = 2 + \sqrt{5}$. For $x > 1$,

$$\log x > \frac{2(x-1)}{x+1},$$

because the difference vanishes at $x = 1$ and has derivative $(x-1)^2/[x(x+1)^2] > 0$. Since $\phi > 11/7$, this gives $a = \log \phi > 4/9$.

For more precise enclosures, let $x > 1$ and $y = (x-1)/(x+1)$. The identity

$$\log x = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + r_N, \qquad 0 < r_N < \frac{2y^{2N+1}}{(2N+1)(1-y^2)} \tag{A.1}$$

follows from the power series for the inverse hyperbolic tangent and a geometric bound on its tail. If x is rational, both the partial sum and the upper remainder bound are rational. Monotonicity of the logarithm also lets us use rational endpoints for algebraic x.

Squaring verifies the enclosure

$$2.2360679774997896964 < \sqrt{5} < 2.2360679774997896965.$$

Applying (A.1) with $N = 60$ to these endpoints and to the corresponding endpoints of ϕ gives rational bounds for a and b. Taking quotients of positive endpoints proves $1.67 < b/a < 1.68$.

Applying the same formula to 2, 3, 5, and 8, and using differences of logarithmic intervals, gives

$$1.29 < 3 - \frac{\log 2}{\log 3 - \log 2} < 1.30,$$

$$1.84 < 4 - \frac{\log 3}{\log 5 - \log 3} < 1.85,$$

$$1.57 < 5 - \frac{\log 5}{\log 8 - \log 5} < 1.58.$$

All denominators stay positive under these enclosures. The accompanying program performs these comparisons with Python's exact `Fraction` arithmetic. Thus the coarse certificates used in the proof are reproducible finite rational computations with analytically controlled tails.

For completeness, the decreasing numerator bound in (5.9) follows from

$$\frac{(7(n+1)/5 - 11/10)s_{n+1}}{(7n/5 - 11/10)s_n} < \frac{2}{5}\frac{14n+3}{14n-11} < 1 \qquad (n \geq 6).$$

The denominator $4/9 - (7/5)s_n$ increases because s_n decreases. These facts justify replacing the full tail in that proof by its value at $n = 6$.

B Reproduction instructions and data inventory

The source archive is organized as follows:

File or directory	Purpose
`fibonacci-logarithms.tex`	Main LaTeX document
`sections/`	Section source files and bibliography
`research.py`	Exact integer generation, checks, and plots
`figures/`	Four figures in PDF and PNG formats
`data/ratios.csv`	High-precision statistics for $3 \leq n \leq 150$
`data/leading_digits.csv`	Digit counts and limiting proportions
`data/verification.json`	Parameters and completed-check metadata
`README.md`	Commands and dependency requirements

To rerun the calculations, install `mpmath`, `numpy`, and `matplotlib`, then run `python3 research.py -output-dir .` from the archive's root. The program stops if an assertion fails. To rebuild the PDF, run `tectonic fibonacci-logarithms.tex`, or use a LaTeX installation supporting the packages in the preamble. The section and figure directories must accompany the main file.

The manuscript tables contain selected values from the generated data, rounded for readability. CSV ratio columns retain 60 significant digits; these are reported values from a 160-digit calculation, not symbolic exact values. Leading-digit counts are exact integers. The plots do not require an internet connection once dependencies are installed. Source articles were consulted for the literature review and theorem context; they are linked below rather than redistributed in the source archive.

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